CHAPTER

23

Image Formation & Display

Images are a description of how a parameter varies over a surface. For example, standard visual images result from light intensity variations across a two-dimensional plane. However, light is not the only parameter used in scientific imaging. For example, an image can be formed of the *temperature* of an integrated circuit, *blood velocity* in a patient's artery, *x-ray emission* from a distant galaxy, *ground motion* during an earthquake, etc. These exotic images are usually converted into conventional pictures (i.e., light images), so that they can be evaluated by the human eye. This first chapter on image processing describes how digital images are formed and presented to human observers.

Digital Image Structure

Figure 23-1 illustrates the structure of a digital image. This example image is of the planet Venus, acquired by microwave radar from an orbiting space probe. Microwave imaging is necessary because the dense atmosphere blocks visible light, making standard photography impossible. The image shown is represented by 40,000 samples arranged in a two-dimensional array of 200 columns by 200 rows. Just as with one-dimensional signals, these rows and columns can be numbered 0 through 199, or 1 through 200. In imaging jargon, each sample is called a **pixel**, a contraction of the phrase: *picture element*. Each *pixel* in this example is a single number between 0 and 255. When the image was acquired, this number related to the amount of microwave energy being reflected from the corresponding location on the planet's surface. To display this as a *visual image*, the value of each pixel is converted into a **grayscale**, where 0 is black, 255 is white, and the intermediate values are shades of gray.

Images have their information encoded in the **spatial domain**, the image equivalent of the time domain. In other words, features in images are represented by *edges*, not *sinusoids*. This means that the spacing and number of pixels are determined by how small of features need to be seen,